

EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

Aim:

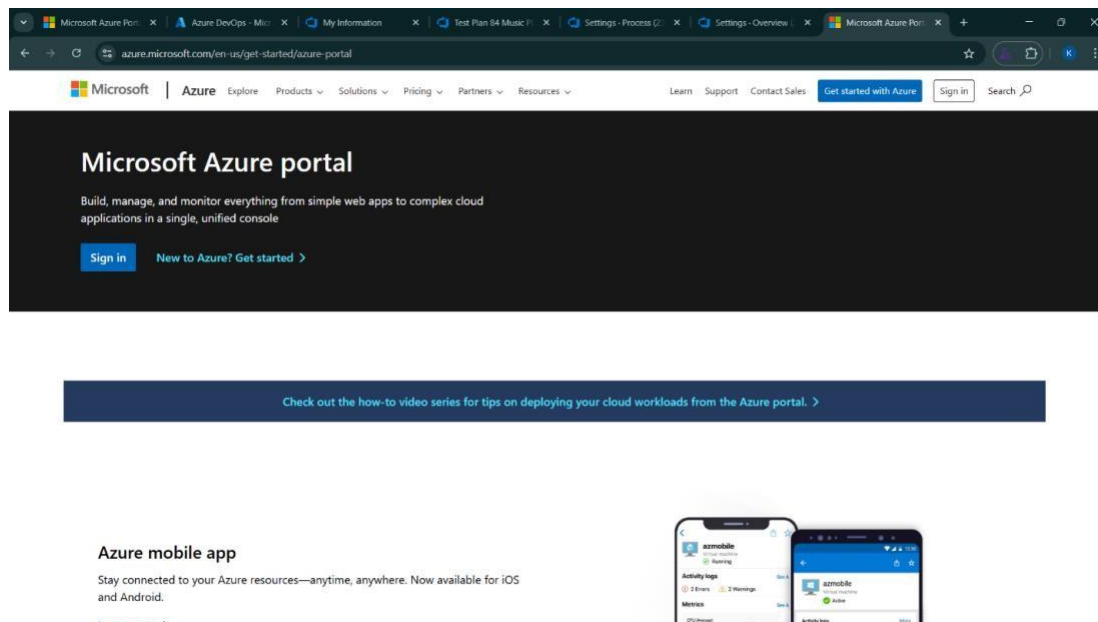
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

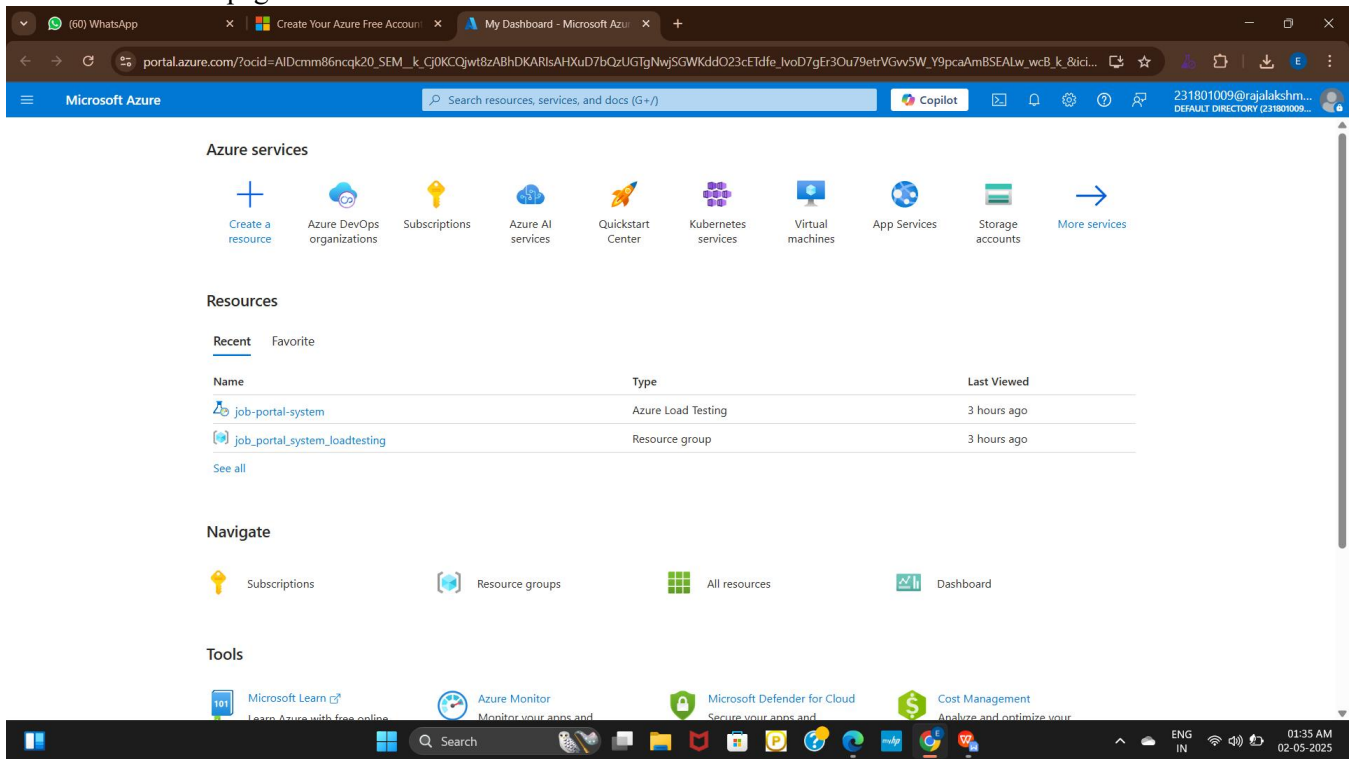
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>

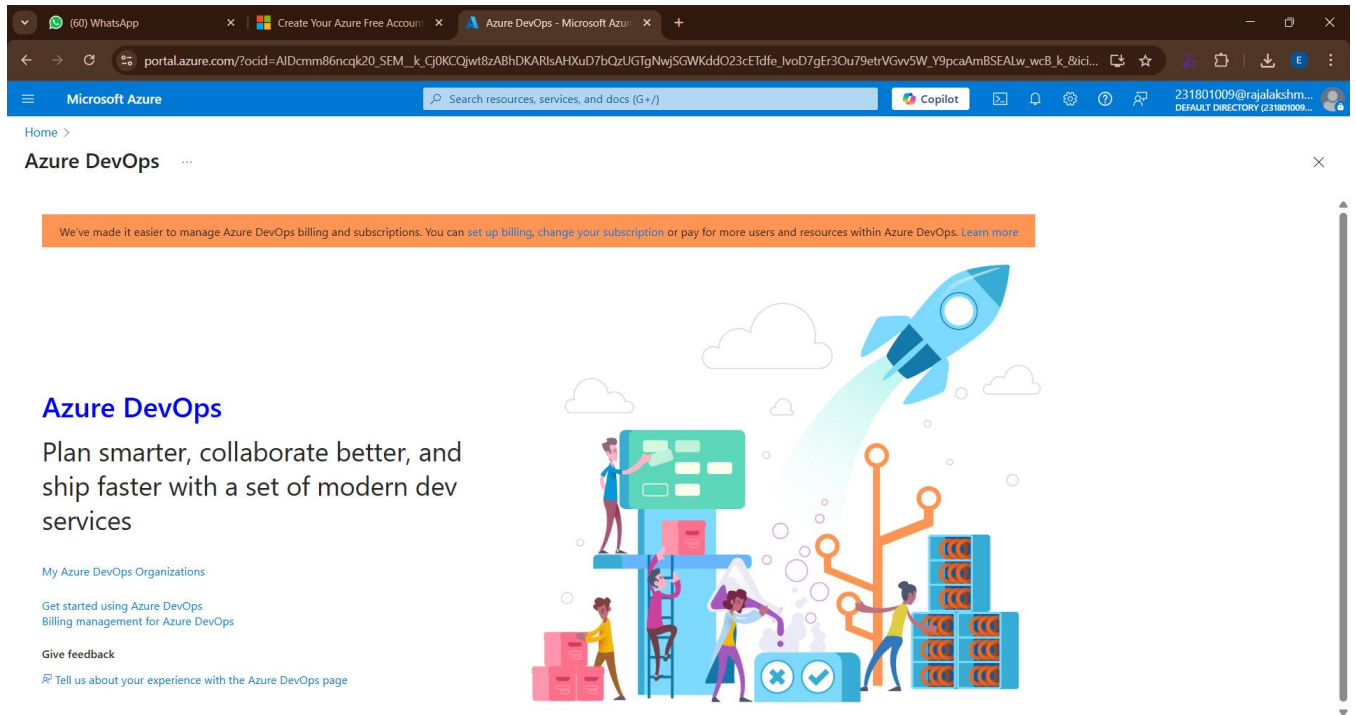


2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.

4. Click on the *My Azure DevOps Organization* link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

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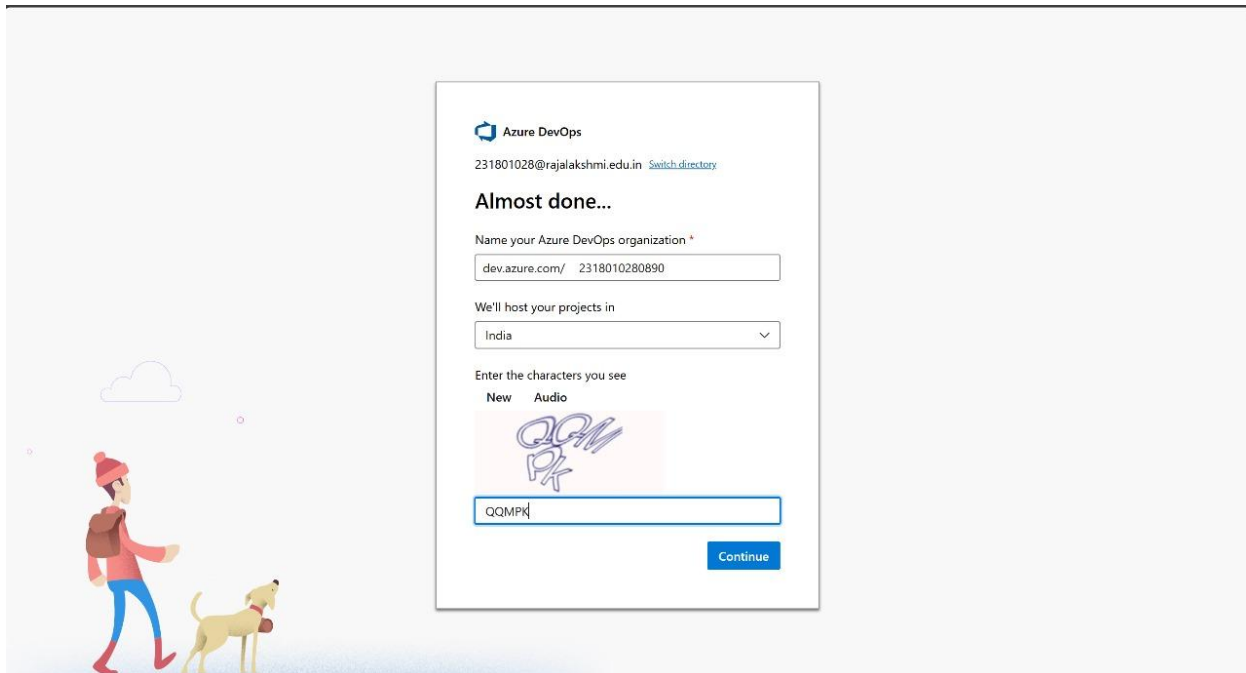
EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account



2. Create the First Project in Your Organization

a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

c. Enter the project name, description, and visibility options:

Name: Choose a name for the project (e.g., **LMS**).

Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

d. Once you've filled out the details, click **Create** to set up your first project.

☰
📁
🔍
👤 DM

Create a project to get started

Project name *

Description

Visibility

Public

Anyone on the internet can view the project. Certain features like TFVC are not supported.

Private

Only people you give access to will be able to view this project.

By creating this project, you agree to the Azure DevOps [code of conduct](#)

^ **Advanced**

Version control ⓘ

Work item process ⓘ

+ Create project

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Microsoft
Deepika M Sign out

Deepika M Edit profile

231801028@rajalakshmi.edu.in

Microsoft account

India
231801028@rajalakshmi.edu.in

Visual Studio Dev Essentials
Get everything you need to build and deploy your app on any platform.
[Use your benefits](#)

Azure DevOps Organizations

[dev.azure.com/231801028](#) (Owner)

▼ [dev.azure.com/Infinitycoders](#) (Owner)

Projects	Actions
Patient Tracker Application New project	Open in Visual Studio

[Create new organization](#)

4. Project dashboard

The screenshot displays the Azure DevOps interface for a project named 'Patient Tracker Application'. The left sidebar contains navigation links: Overview, Summary, Dashboards, Wiki, Boards, Repos, Pipelines, Test Plans, and Artifacts. The main content area is titled 'Patient Tracker Application' and includes a search bar, a 'Private' status indicator, and an 'Invite' button. The dashboard is divided into three main sections: 'About this project', 'Project stats', and 'Members'.

About this project

Help others to get on board!
Describe your project and make it easier for other people to understand it.

+ Add Project Description

Project stats (Period: Last 7 days)

Boards

- 0 Work items created
- 0 Work items completed

Pipelines

- 0% Builds succeeded

Members (4)

- DM, HU, AT, AM

5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot displays the Azure Boards interface for a project named "Patient Tracker Application Team". The left-hand navigation menu is visible, with "Boards" selected. The main area shows the "Backlog" view, which contains a list of work items. The work items are categorized as "Epic" and include titles such as "User Management & Authentication", "Patient Record Management", "Monitoring & Notifications", and "Admin Tools, Logging & Reporting". The state of all items is "New".

Order	Work Item Type	Title	State	Effort	Busin...	Value Area	Tags
1	Epic	> 🏰 User Management & Authentication	🟡 New			Business	
2	Epic	> 🏰 Patient Record Management	🟡 New			Business	
3	Epic	> 🏰 Monitoring & Notifications	🟡 New			Business	
4	Epic	> 🏰 Admin Tools, Logging & Reporting	🟡 New			Business	

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

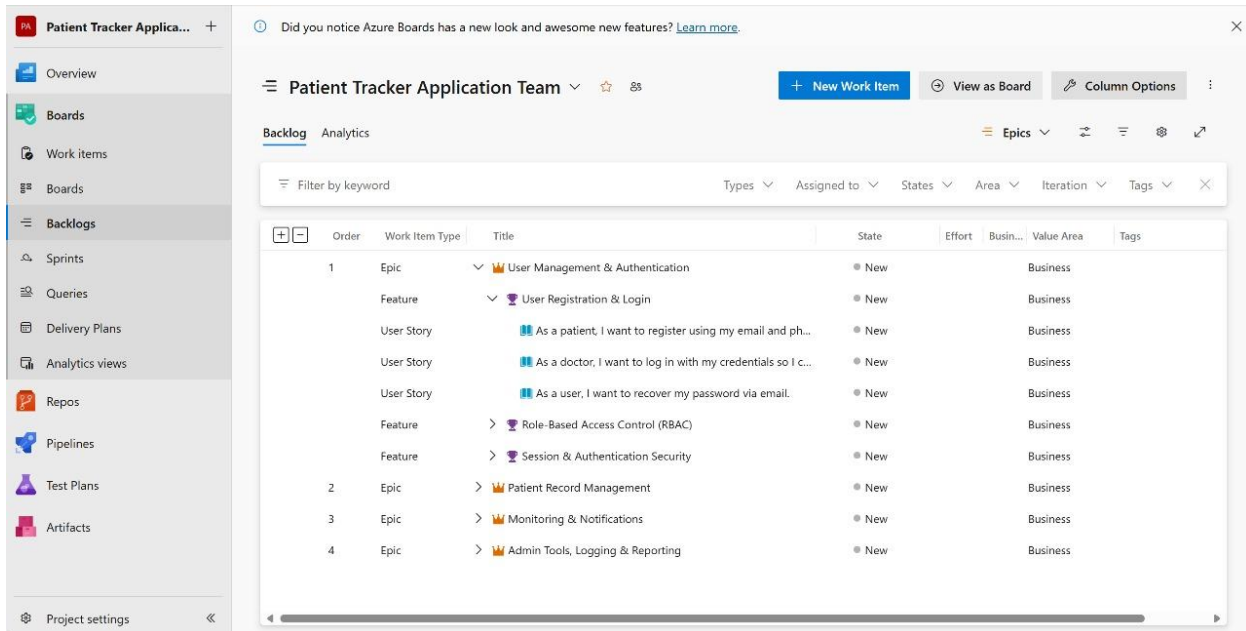
EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

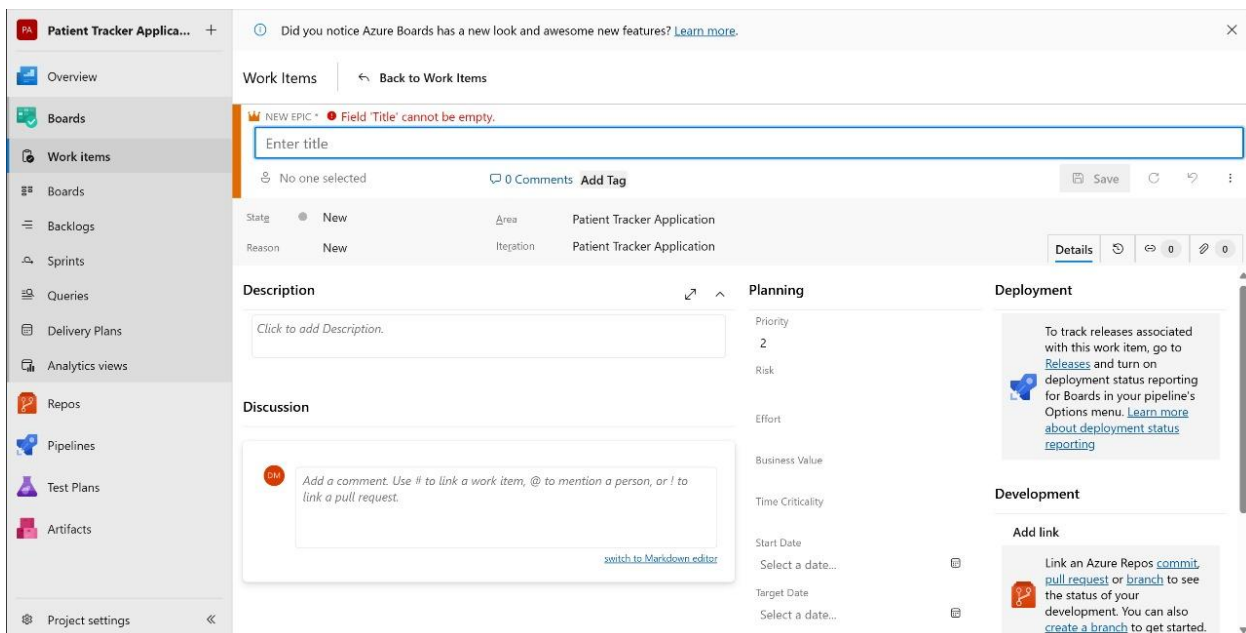
Create Epic, Features, User Stories, Task



The screenshot shows the Azure Boards interface for the 'Patient Tracker Application Team'. The left sidebar contains navigation options: Overview, Boards, Work items, Backlogs (selected), Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area displays the 'Backlog' view with a table of work items. The table has columns for Order, Work Item Type, Title, State, Effort, Business Value, and Tags. The work items are organized into four epics: 1. User Management & Authentication (containing 6 items), 2. Patient Record Management (1 item), 3. Monitoring & Notifications (1 item), and 4. Admin Tools, Logging & Reporting (1 item). All items are in the 'New' state.

Order	Work Item Type	Title	State	Effort	Business Value	Tags
1	Epic	User Management & Authentication	New		Business	
	Feature	User Registration & Login	New		Business	
	User Story	As a patient, I want to register using my email and ph...	New		Business	
	User Story	As a doctor, I want to log in with my credentials so I c...	New		Business	
	User Story	As a user, I want to recover my password via email.	New		Business	
	Feature	Role-Based Access Control (RBAC)	New		Business	
	Feature	Session & Authentication Security	New		Business	
2	Epic	Patient Record Management	New		Business	
3	Epic	Monitoring & Notifications	New		Business	
4	Epic	Admin Tools, Logging & Reporting	New		Business	

1.Fill in Epics



The screenshot shows the 'Work Items' view in Azure Boards for the 'Patient Tracker Application' team. A new work item is being created. The 'Title' field is empty, and a message states 'Field Title cannot be empty.' The 'Statg' is set to 'New', and the 'Reason' is also 'New'. The 'Area' is 'Patient Tracker Application' and the 'Iteration' is also 'Patient Tracker Application'. The 'Description' field is empty, with a placeholder 'Click to add Description.' The 'Discussion' section is empty, with a placeholder 'Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.' The 'Planning' section shows 'Priority' as 2, 'Risk' as 0, 'Effort' as 0, and 'Business Value' as 0. The 'Deployment' section has a note about tracking releases. The 'Development' section has a note about linking to Azure Repos.

Work Items | Back to Work Items

NEW EPIC • Field 'Title' cannot be empty.

Enter title

No one selected | 0 Comments | Add Tag

Save | Cancel | Refresh | Help

Statg: New | Area: Patient Tracker Application | Reason: New | Iteration: Patient Tracker Application

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[switch to Markdown editor](#)

Planning

Priority: 2

Risk: 0

Effort: 0

Business Value: 0

Time Criticality: 0

Start Date: Select a date...

Target Date: Select a date...

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

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2. Fill in Features

Patient Tracker Application +

Overview

Boards

Work Items

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Did you notice Azure Boards has a new look and awesome new features? [Learn more.](#)

Work Items | [Back to Work Items](#)

NEW FEATURE * Field 'Title' cannot be empty.

Enter title

No one selected 0 Comments [Add Tag](#) [Save](#) [Refresh](#) [Share](#)

Status	Area	Reason
New	Patient Tracker Application	New
	Iteration	Patient Tracker Application

Description [Link](#) [Up](#)

Click to add Description.

Discussion

[Add comment](#)

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[Switch to Markdown editor](#)

Planning

Priority
2

Risk

Effort

Business Value

Time Criticality

Start Date
Select a date...

Target Date
Select a date...

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

3.Fill in User Story Details

Did you notice Azure Boards has a new look and awesome new features? [Learn more.](#)

Work Items | [Back to Work Items](#)

NEW USER STORY • ● Field 'Title' cannot be empty.

Enter title

No one selected | 0 Comments | Add Tag

State	New	Area	Patient Tracker Application
Reason	New	Iteration	Patient Tracker Application

Description

Click to add Description.

Acceptance Criteria

Click to add Acceptance Criteria.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Planning

Story Points

Priority

2

Risk

Classification

Value area

Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4

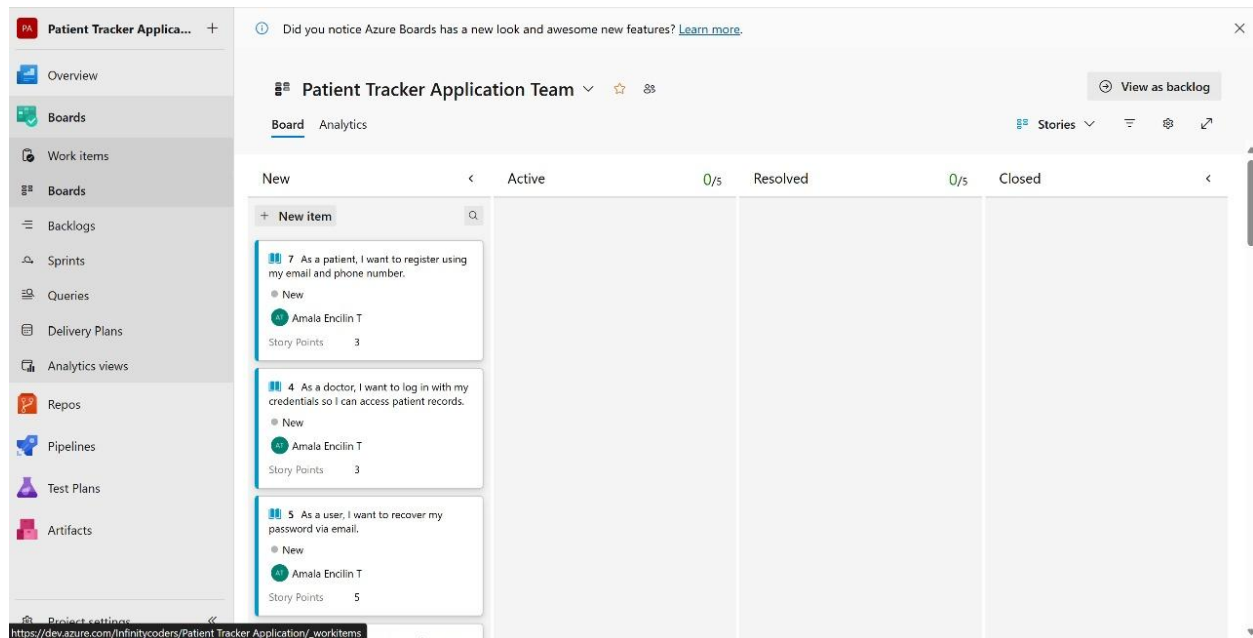
SPRINT PLANNING

Aim:

To assign user story to specific sprint for patient tracker application.

Sprint Planning

Sprint



Result:

The Sprints are created for the patient tracker application.

EXP NO: 5

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories -Patient tracker application.

Poker Estimation

USER STORY 7

7 As a patient, I want to register using my email and phone number.

Amala Encilin T

0 Comments Add Tag

Save and Close

Follow

Updated by Deepika M: 4 Apr

Details

1

0

Description

Click to add Description.

Acceptance Criteria

1. The system must require mandatory fields (name, age, gender, contact, medical history).

2. Each patient should have a unique **Patient ID**, preventing duplicate entries.

3. Only **doctors** can add new patient records; unauthorized users get an error.

4. A success message should appear after saving, and the patient should be visible in the list.

5. The record should save within **2 seconds**, with an error message if it fails.

6. The system should log **who added the record and when** for tracking.

Planning

Story Points

3

Priority

2

Risk

Classification

Value area

Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Discussion

Result:

The Estimation/Story Points is created for the project using Poker Estimation.

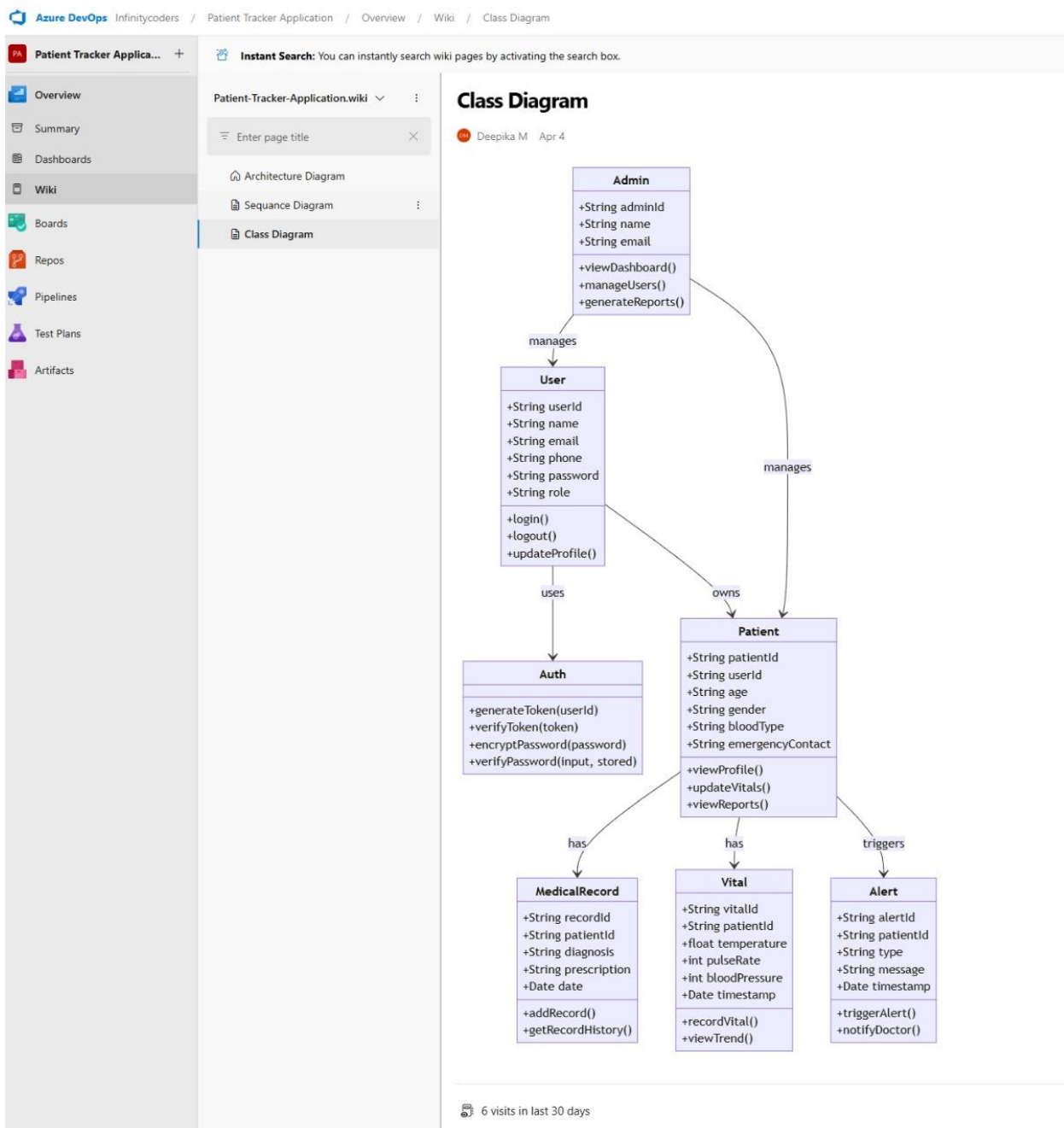
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

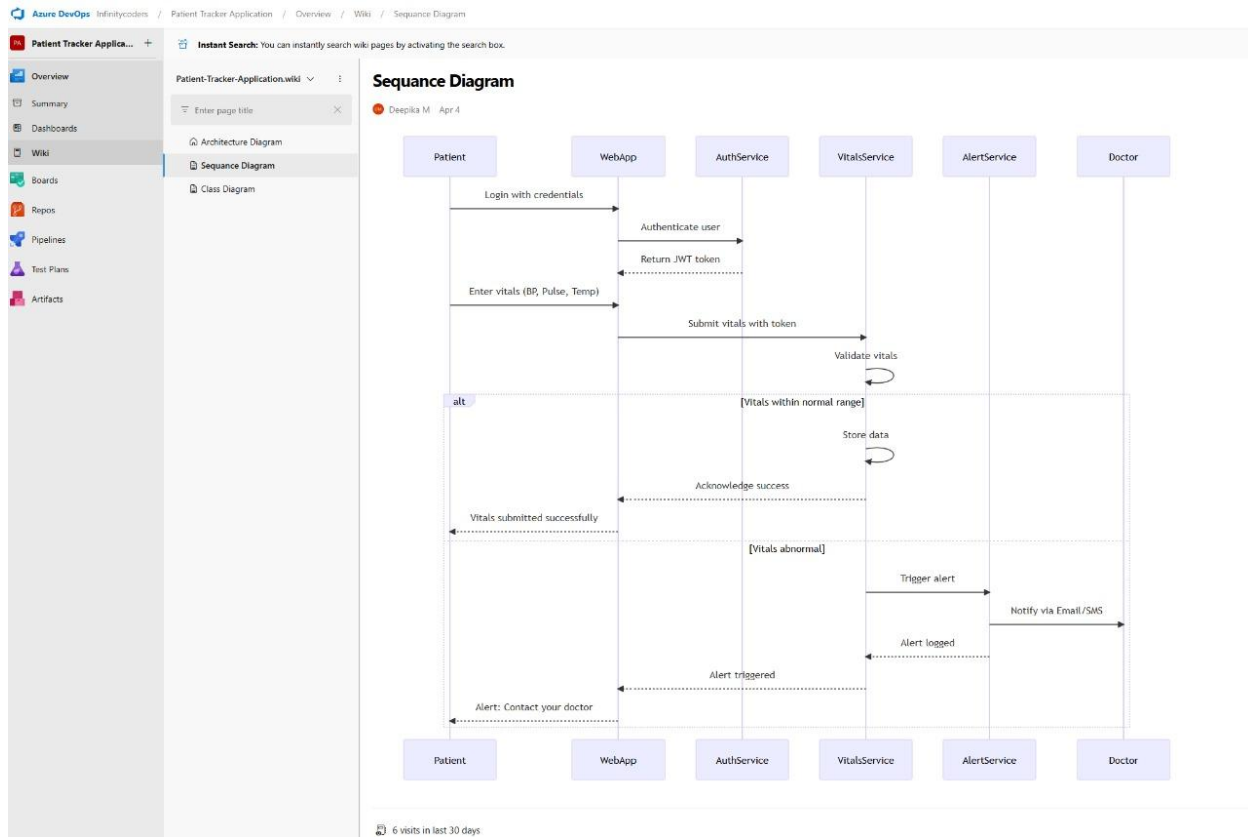
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the patient tracker application.

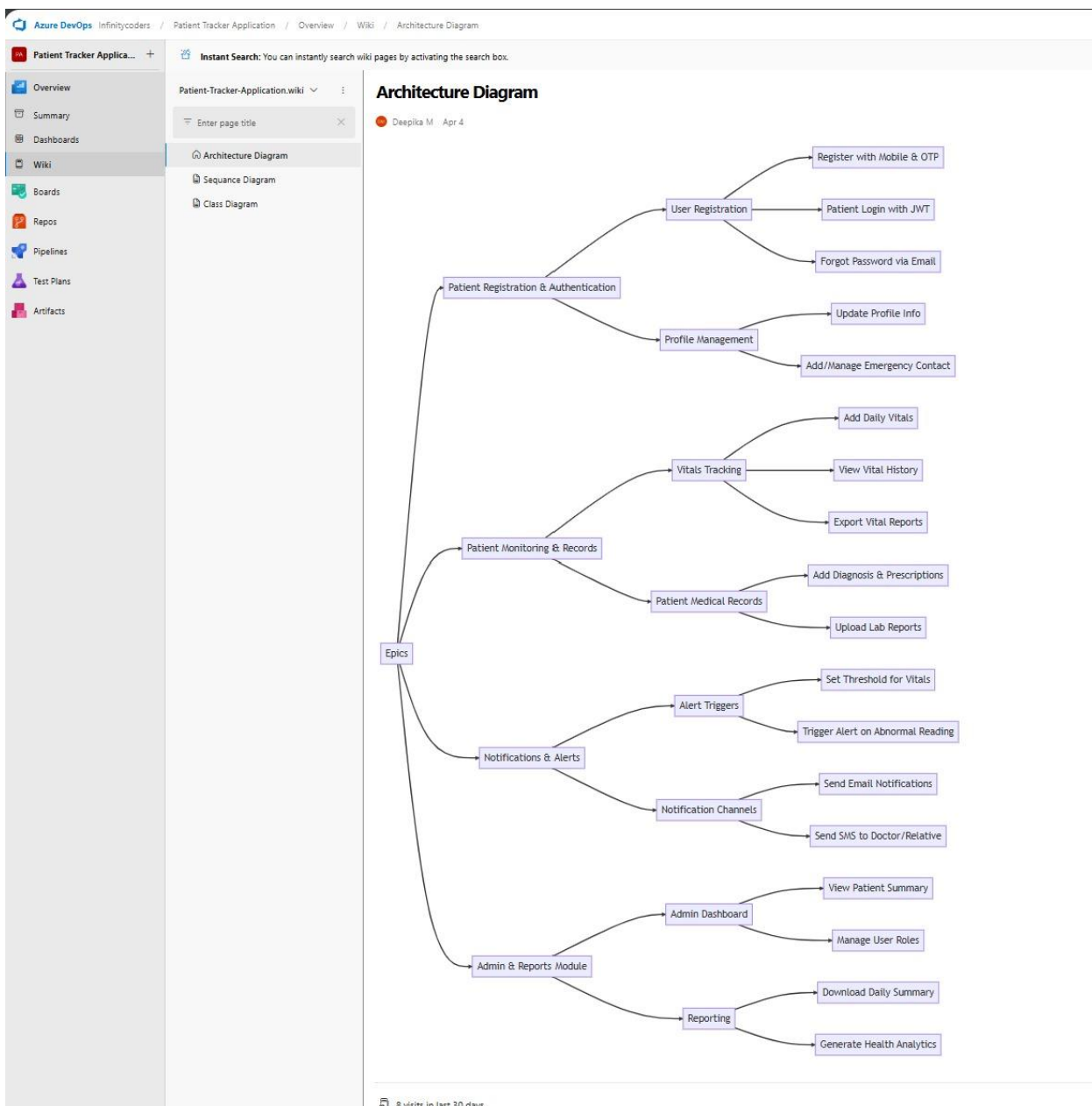
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

Architectural Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the patient tracker application.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
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Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Viewing and Managing Playlists
- Fetching Real-time Metadata
- Editing playlists (rename, reorder, record)
- Creating smart audio playlists based on categories (mood, genre, artist, etc.)

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly (e.g., TC01 – Successful Login, TC10 – Save Playlist Fails).
- Helps in quick identification and linking to user stories or features.

7. Separate Test Suites

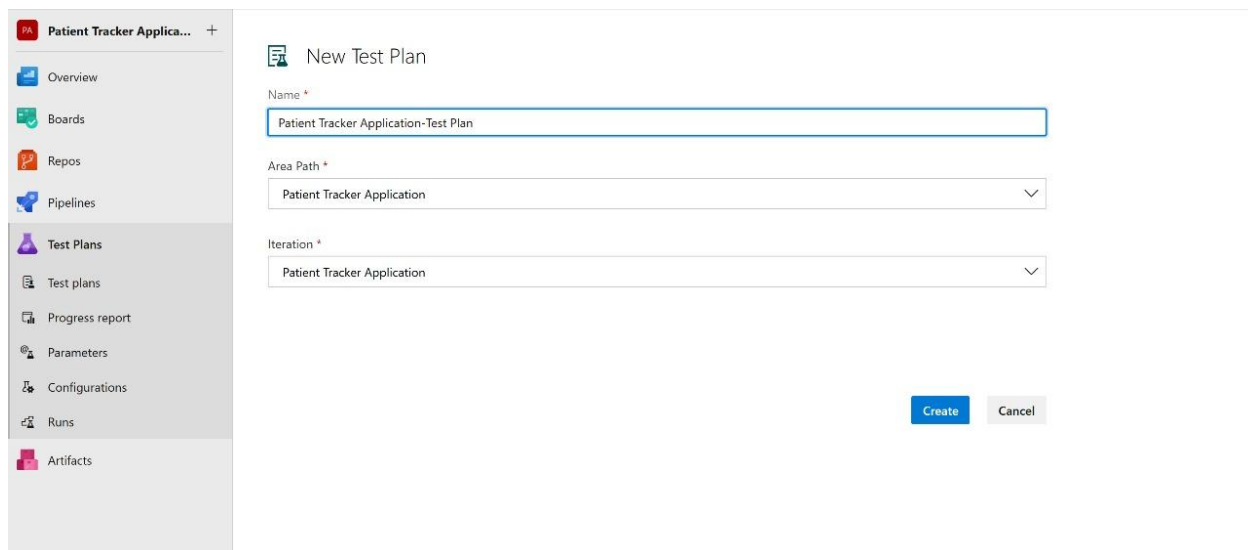
- Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).

- Improves organization and test execution flow in Azure DevOps.

8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

1. New test plan



New Test Plan

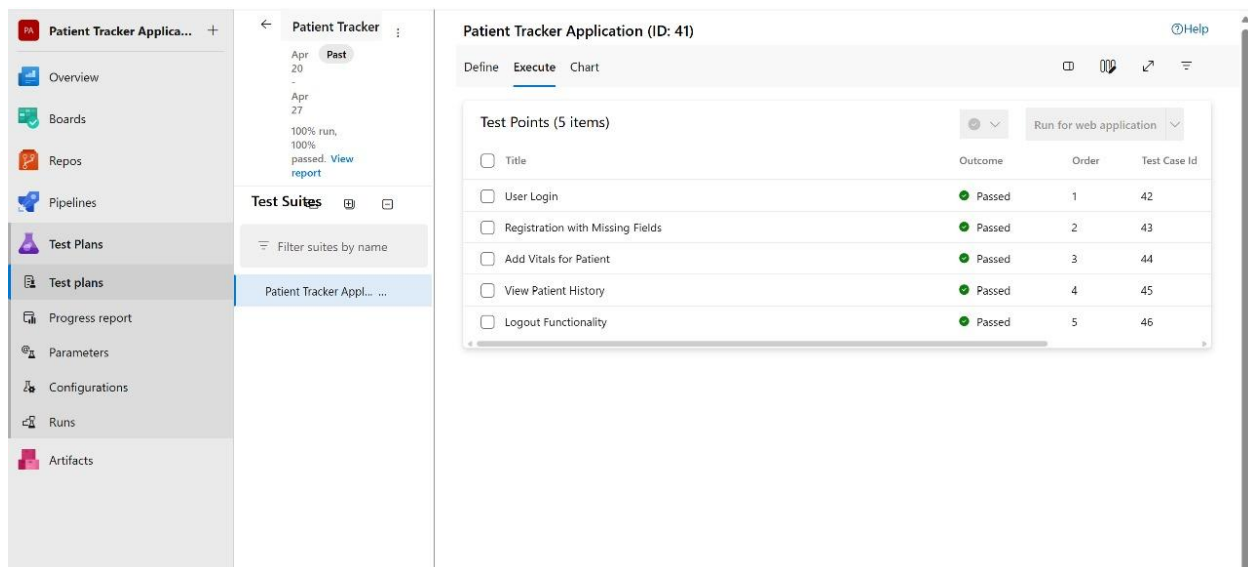
Name *
Patient Tracker Application-Test Plan

Area Path *
Patient Tracker Application

Iteration *
Patient Tracker Application

Create Cancel

2. Test suite



Patient Tracker Application (ID: 41)

Define Execute Chart

Test Points (5 items)

Title	Outcome	Order	Test Case Id
User Login	Passed	1	42
Registration with Missing Fields	Passed	2	43
Add Vitals for Patient	Passed	3	44
View Patient History	Passed	4	45
Logout Functionality	Passed	5	46

3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Batch Creator – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access my playlists (ID: 79).
- As a user, I need to see my playlist in one place (ID: 76).
- As a user, I should be able to create an audio playlist as needed (ID: 73).
- As a user, I should be able to rename, record, and change the playlist (ID: 68).
- As a user, I need to have real-time metadata (ID: 65).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. TC01 – Successful Sign Up

- **Action:**
 - Go to the Sign-Up page.
 - Enter valid name, email, and password.
 - Click "Sign Up".
- **Expected Results:**
 - Sign-Up form is displayed.
 - Fields accept values without error.
 - Account is created, and the user is redirected to the dashboard.
- **Type:** Happy Path

2. TC02 – Secure Login

- **Action:**
 - Go to the Login page.
 - Enter valid email and password.
 - Click on "Login".
- **Expected Results:**
 - Login form is displayed.
 - Fields accept data without error.
 - User is logged in and redirected to the dashboard.
- **Type:** Happy Path

3. TC03 – Sign Up with Existing Email

- **Action:**
 - Go to the Sign-Up page.
 - Enter a name and an already registered email.
 - Click on "Sign Up".
- **Expected Results:**

- Fields accept data.
- Error message "Email already registered" is displayed.
- **Type:** Error Path

4. TC04 – Login with Wrong Password

- **Action:**
 - Go to the Login page.
 - Enter valid email and incorrect password.
 - Click on "Login".
- **Expected Results:**
 - Input is accepted.
 - Error message "Invalid username or password" is shown.
- **Type:** Error Path

Test Suit: TS02 - View Playlists (ID: 87)

1. TC05 – View Playlist Page

- **Action:**
 - Log in successfully.
 - Navigate to "My Playlists" section.
- **Expected Results:**
 - All created playlists are displayed clearly.
- **Type:** Happy Path

2. TC06 – Playlist Loading Failure

- **Action:**
 - Disconnect from the internet.
 - Navigate to "My Playlists".
- **Expected Results:**
 - Network is offline.
 - Error message "Unable to load playlists" is shown.
- **Type:** Error Path

Test Suit: TS03 - Real-Time Metadata (ID: 88)

1. TC07 – Real-Time Metadata Display

- **Action:**
 - Play a song.
 - Observe the metadata panel.
- **Expected Results:**
 - Metadata (title, artist, album, duration) is displayed and updates in real time.
- **Type:** Happy Path

2. TC08 – Metadata Not Updating

- **Action:**

- Play a different song.
 - Observe the metadata panel.
- **Expected Results:**
 - Metadata remains static or shows default/fallback message.
- **Type:** Error Path

Test Suit: TS04 - Playlist Editing (ID: 89)

1. TC09 – Rename Playlist Successfully

- **Action:**
 - Navigate to "My Playlists".
 - Click "Rename" next to a playlist.
 - Enter a new name and click "Save".
- **Expected Results:**
 - Playlist name updates successfully.
- **Type:** Happy Path

2. TC10 – Rename with Blank Name

- **Action:**
 - Click "Rename" on a playlist.
 - Leave the field blank.
 - Click "Save".
- **Expected Results:**
 - Error message "Playlist name cannot be empty" is shown.
- **Type:** Error Path

3. TC11 – Change Playlist Order

- **Action:**
 - Open a playlist.
 - Drag and drop songs to reorder.
 - Click "Save".
- **Expected Results:**
 - Playlist order is updated and saved.
- **Type:** Happy Path

4. TC12 – Change Playlist Order Fails

- **Action:**
 - Login and go to "My Playlists".
 - Select a playlist.
 - Go offline or simulate server error.
 - Reorder songs and click "Save Order".
- **Expected Results:**
 - Error message: "Failed to update order. Please check your connection".
- **Type:** Error Path

Test Suit: TS05 - Smart Playlist Creation (ID: 90)

1. TC13 – Generate Playlist Based on Various Categories

- **Action:**
 - Login with valid credentials.
 - Click on "Generate Playlist".
 - Select categories.
 - Click "Generate Playlist".
- **Expected Results:**
 - Playlist is generated based on selected mood and categories.
- **Type:** Happy Path

2. TC14 – Fail to Generate Playlist Due to Missing Category Selection or Invalid Input

- **Action:**
 - Login with valid credentials.
 - Click on "Generate Playlist".
 - Select categories.
 - Click "Generate Playlist".
- **Expected Results:**
 - Error message: "Please select at least one valid category" or "No recommendations found for the selected filters".
- **Type:** Error Path

Test Cases

NEW TEST CASE *

User Login

Deepika M 0 Comments Add Tag Save and Close

State Design Area Patient Tracker Application Reason New Iteration Patient Tracker Application Steps Summary Associated Automation

Steps

Steps	Action	Expected result
1.		
2.		

Click or type here to add a step

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

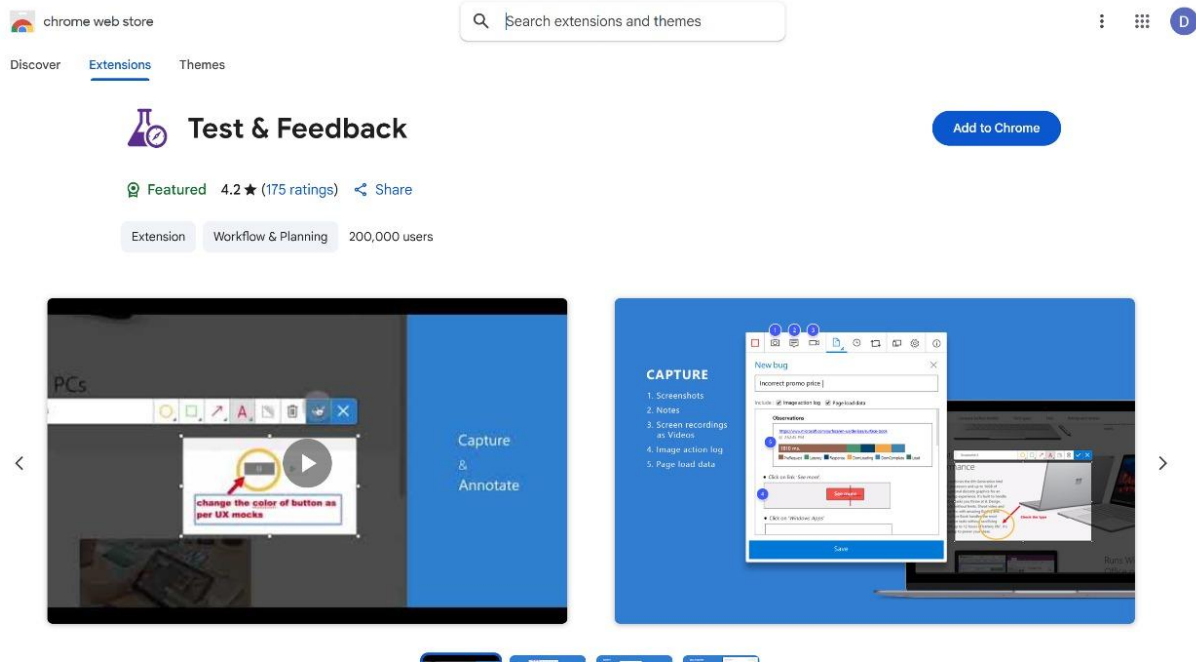
Related Work

Add link

[Add an existing work item as a parent](#)

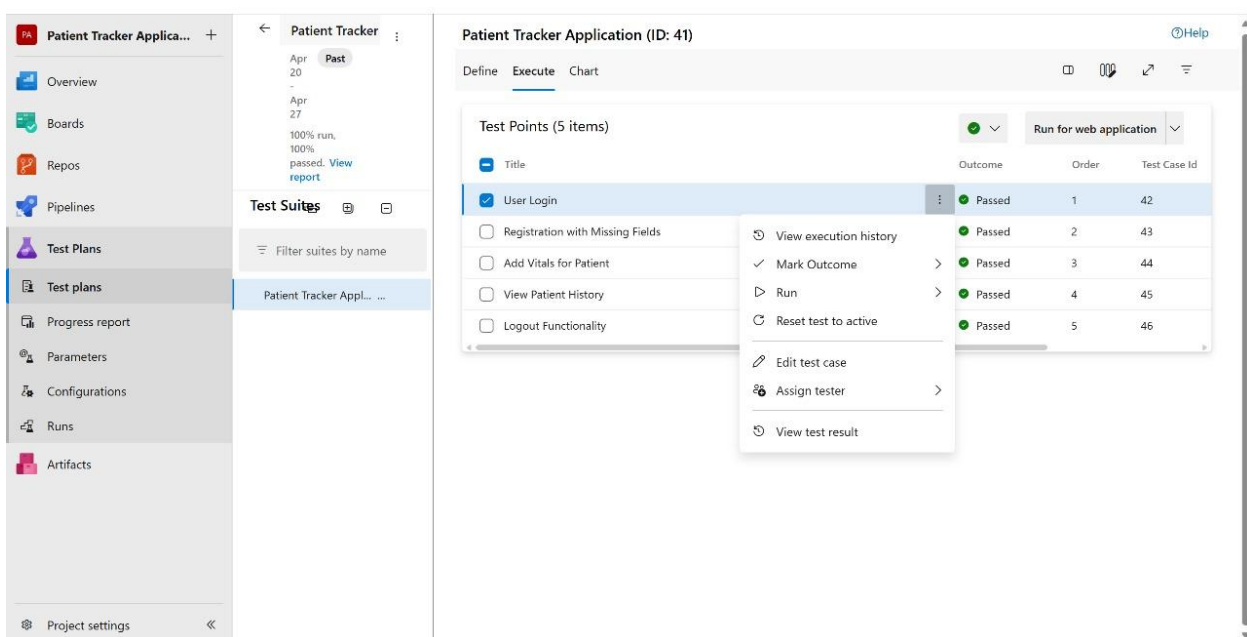
Parameter values

4. Installation of test



Test and feedback
Showing it as an extension

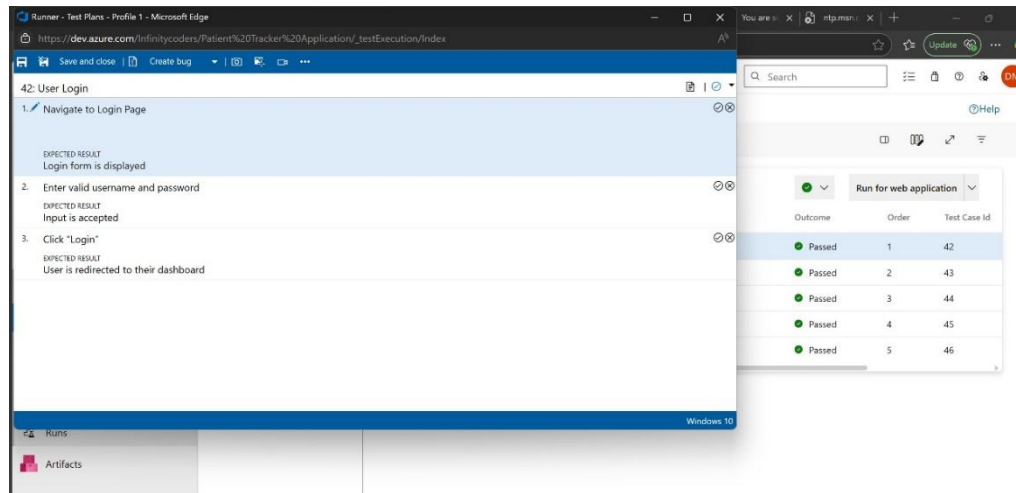
5. Running the test cases



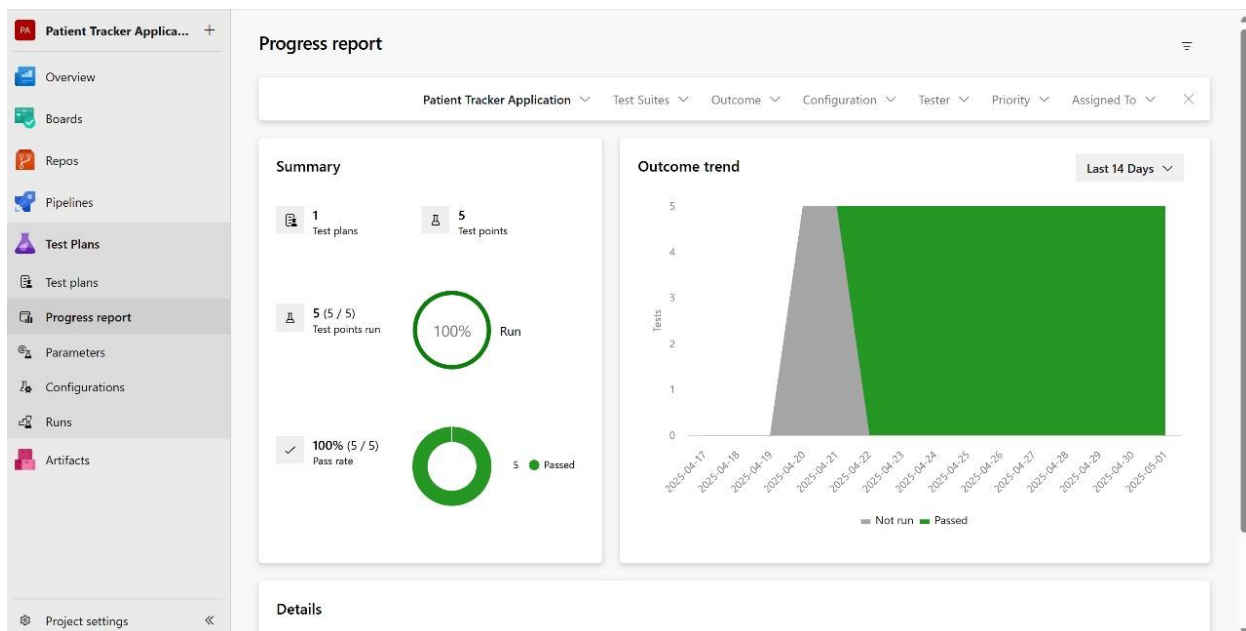
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Recording test case



Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9

LOAD TESTING AND PERFORMANCE TESTING

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint.

Load Testing

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

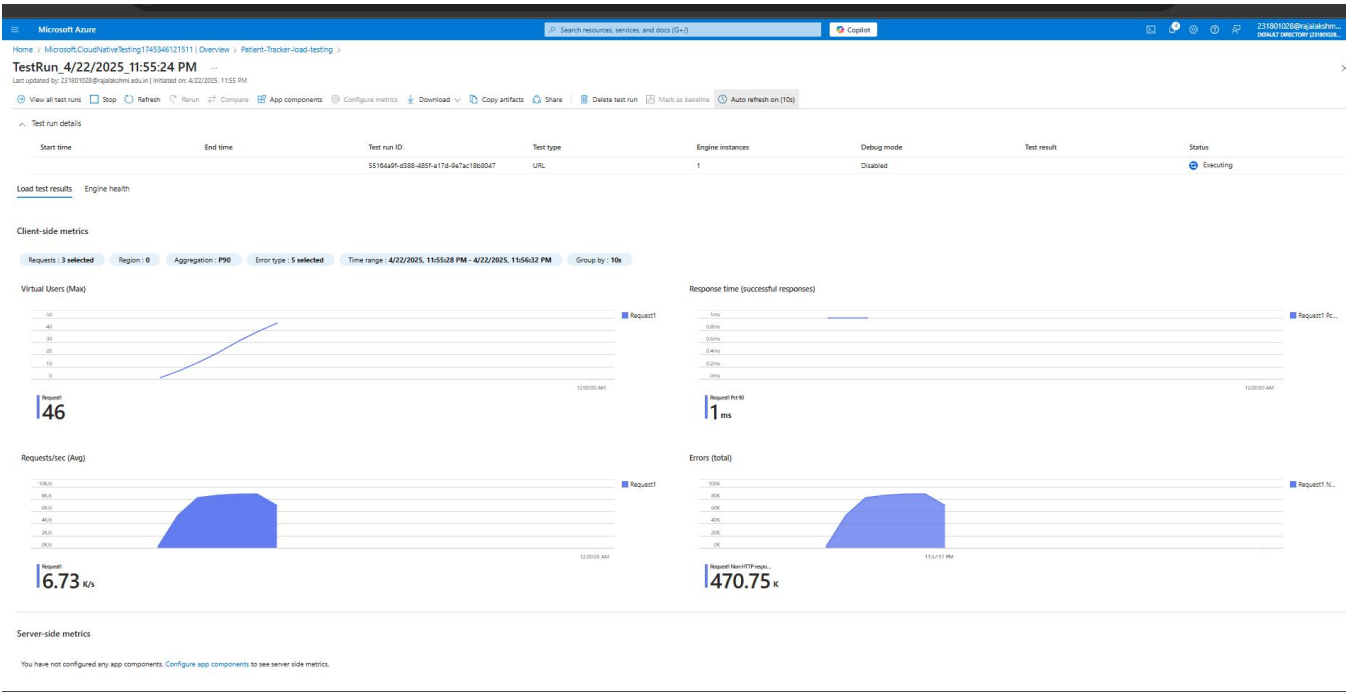
1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - Go to *Create a resource* → Search for “Azure Load Testing”.
 - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - *Subscription*: Choose your Azure subscription.
 - *Resource Group*: Create new or select an existing one.
 - *Name*: Provide a unique name (no special characters).
 - *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - *Test Name*: Provide a unique name.
 - *Description*: (Optional) Add test purpose.
 - *Run After Creation*: Keep checked.
3. Load Settings
 - *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing



+ Add New Patient

Patient Name

Condition / Note

Add Patient

Dashboard

Total Patients: 58

Upcoming Appointments: 12

Alerts: 3

Patient Records

John Doe

Condition: Hypertension | Last Visit: 12-Apr-2025

Jane Smith

Condition: Diabetes Type 2 | Last Visit: 18-Apr-2025

Rahul Khanna

Condition: Asthma | Last Visit: 10-Apr-2025

Pipeline:

PA Patient Tracker Applica... +

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

Project settings <<

Pipelines

Recent All Runs

Filter pipelines

New pipeline

Recently run pipelines

Pipeline	Last run
✖ Deepika26072006.Patient_Tracker_App...	#20250423.1 • Set up CI with Azure Pipelines Individual CI for main Apr 23 <1s

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Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint.

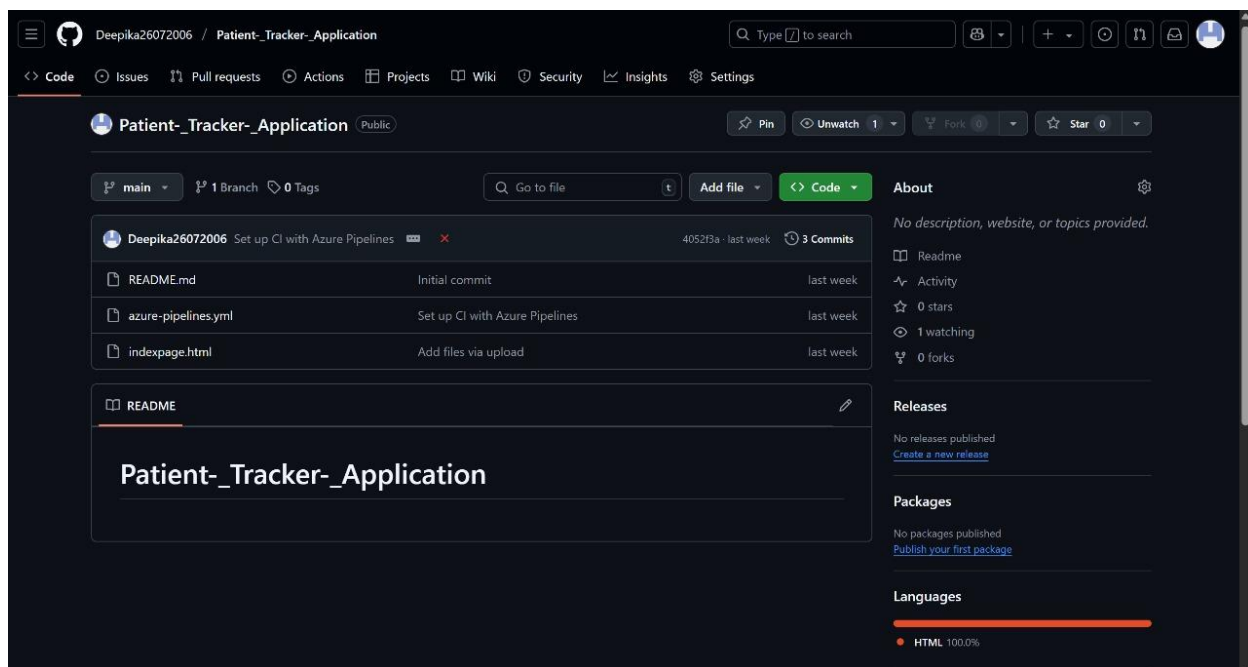
EXP NO: 10

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Patient tracker application.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.